Independent audit of the Toda–Boussinesq frame dictionary package: endpoint identities, dispersion budgets, mixed remainders, and two-body frame constraints

Quantum Boussinesq solitons project

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Abstract

We report an independent mathematical and numerical audit of the `frame_dictionary` package (`phase2/frame_dictionary.tex`, `frame_identities.py`, `frame_edge.py`, `frame_analysis.py`, and their JSON outputs), which addresses ranked priority #1 of the Phase 2 master report: the common quantum momentum and frame dictionary between the quantum Toda Fermi-edge limit and continuum Boussinesq solitons at fixed physical observables. **(c)** The finite-sea Fredholm identities (dressed-energy endpoint identity, thermodynamic pressure response, exact quantum momentum Jacobian, and four-term endpoint reciprocity) are confirmed analytically and verified to 70 decimal digits using an independent Fejér-I (Chebyshev-root) Nyström solver. We also derive a new exact finite-sea opposite-edge dispersion identity and supply direct numerical y-derivative checks of both Jacobian equations in Eq. (7) of `frame_dictionary.tex`, filling a test-coverage gap in the package. **(c)** An independent Fejér-I dispersion scan on $h \in \{0.06, 0.03, 0.015, 0.0075\}$ with Neville and rational Padé [2/1] extrapolations confirms the central entries of Tables 1–2, while qualifying their error budgets and asymptotic identification: (i) at $y = 0.1$, the package's propagated mesh diagnostic for $c_E(0.1)$ exceeds its edge-fit spread and is dominated by float64 cancellation and root-finding tolerance amplified by $1/(hP_0) \simeq 1.04 \times 10^7$; (ii) at $y = 64$, the four-point fits include $h = 0.04$ where $\kappa \simeq 2.05$ exits the small-amplitude regime; and (iii) the sound-speed shift $(v_s - b)/h$ equals the exact analytical constant $1/(4\pi)$, while $c_E(y)$ interpolates between $c_E(0^+) = 1/(4\pi)$ and $c_E(\infty) = 1/(6\pi)$ with two-loop tail $-d_0/(2y)$. **(c)** The survival of the mixed classical-frame remainder $\mathcal{D}_h(Y)$ as $h \to 0$ is confirmed across five alternative extrapolation bases and three new point sets; we further prove that in any mixed rectangle the additive endpoint subtraction terms cancel identically, reducing $\mathcal{D}_0(Y)$ analytically to a pure function of the one-body Jacobian $u'(y)$ with exact asymptote $\lim_{Y\to\infty} \mathcal{D}_0(Y) = -(15\sqrt{2} - 20)/\pi \simeq -0.38617462$. **(c)** /**(d)** Finally, the two-body antisymmetric counterexample $A_\lambda(y_1, y_2)$ in Eq. (18) is confirmed as an algebraic ambiguity of the incomplete subset of constraints tested by the package (one-body dispersion and symmetric reciprocity), but is refuted as a physical ambiguity of the Toda/Boussinesq problem: it is excluded by spatial cluster decomposition (which restricts Darboux frame shifts to finite-rank bilinears of one-body charges, whereas A_λ has infinite Schmidt rank), by the microscopic Krein–Wiener–Hopf differential equation, and by rapidity-ratio reflection parity and analyticity.

1 Scope, conventions, and reproducibility

Labels follow the repository ground rules in `AGENTS.md`: **(a)** proved in the literature, **(b)** conjectured or heuristic in the literature, **(c)** derived or verified in this project, and **(d)** conjecture or open physical identification. No external references are invented or newly claimed to have been read; we rely only on the repository sources `README.md`, `phase2/report.tex`,

`phase2/frame_dictionary.tex`, `phase2/audit_toda_dictionary.tex`, `phase2/independent_phase2h_audit.tex`, and their associated scripts and JSON outputs.

All independent derivations and numerical checks are implemented in `phase2/scripts/audit_frame_dictionary.py` with results archived in `phase2/scripts/audit_frame_dictionary_results.json`. None of the package's solver classes, root finders, or extrapolation routines is imported or reused. Our script records and verifies the SHA-256 hashes of all seven audited files before and after execution, confirming that no existing file is modified.

We retain the fixed Boussinesq conventions of `AGENTS.md` and the inherited Toda–Boussinesq scaling dictionary of `frame_dictionary.tex` Eq. (1):

$$h = \hbar_T = 3b\hbar_B, \quad t = 2\pi h, \quad b = e^{-a/2}, \quad \kappa = \sqrt{3}\beta = \sqrt{yh/b},$$

with classical Toda soliton momentum and energy

$$P_0(\kappa) = \frac{4b}{a}(\kappa\cosh\kappa - \sinh\kappa), \qquad E_0(\kappa) = 2b^2(\sinh\kappa\cosh\kappa - \kappa).$$

Script execution and runtimes (Point 6). All three package scripts and our independent audit script execute cleanly from the repository root and terminate by printing `ALL CHECKS PASSED`. **(c)** Table 1 records the measured wall-clock runtimes on this machine alongside the full-recompute timing archived in `frame_edge_data.json`.

Script	Measured (s)	Recorded JSON (s)	Terminal status
`phase2/scripts/frame_identities.py`	140.18	177.77	`ALL CHECKS PASSED`
`phase2/scripts/frame_edge.py` (cached)	1.62	325.81 (full scan)	`ALL CHECKS PASSED`
`phase2/scripts/frame_analysis.py`	0.19	0.031	`ALL CHECKS PASSED`
`phase2/scripts/audit_frame_dictionary.py`	173.68	173.68	`ALL CHECKS PASSED`

Table 1: Measured and archived wall-clock runtimes of the audited package scripts and the independent audit script (**(c)** , Point 6).

2 Point 1: Re-derivation of the dressed-energy endpoint identity and quantum Jacobian

2.1 Analytic re-derivation from the finite-sea Bethe equations

On $[-q, q]$, let $\Theta(d) = 2h\,\mathrm{Im}\log\Gamma(1+id/h)+2d\log h$ and $K(d) = \Theta'(d) = 2\,\mathrm{Re}\,\psi(1+id/h)+2\log h$. Since $\Theta(-d) = -\Theta(d)$ and $K(-d) = K(d)$, the finite-sea integral operator

$$(\mathcal{A}f)(x) = tf(x) - \int_{-q}^{q} K(x-z)f(z)\,\mathrm{d}z$$

is real, symmetric, and parity-preserving on $L^2[-q, q]$. With $\mathcal{A}\rho = 1$, $\mathcal{A}\eta = x$, $\int_{-q}^{q}\rho(x)\,\mathrm{d}x = 1/a$, and $\mathcal{A}F_p(x) = \Theta(p-x)$, self-adjointness immediately yields **(c)**

$$P(p) = p + \langle 1, F_p\rangle = p + \langle \mathcal{A}\rho, F_p\rangle = p + \int_{-q}^{q}\Theta(p-x)\rho(x)\,\mathrm{d}x, \tag{1}$$

$$E(p) = \frac{p^2}{2} + \langle x, F_p\rangle = \frac{p^2}{2} + \langle \mathcal{A}\eta, F_p\rangle = \frac{p^2}{2} + \int_{-q}^{q}\Theta(p-x)\eta(x)\,\mathrm{d}x. \tag{2}$$

Differentiating in p gives $P'(p) = 1 + \int_{-q}^{q}K(p-x)\rho(x)\,\mathrm{d}x =: D(p)$ and $E'(p) = p + \int_{-q}^{q}K(p-x)\eta(x)\,\mathrm{d}x =: V(p)$. Evaluating the sea equations at $x = \pm q$ gives $D(q) = t\rho(q)$ and $V(q) = t\eta(q)$, while integrating $P'(p) = t\rho(p)$ across $[-q, q]$ using oddness of P gives $P(\pm q) = \pm\pi h/a$. **(c)**

To re-derive the boxed endpoint identities in Eq. (5) of `frame_dictionary.tex`, differentiate $(\mathcal{A}F_p)(x) = \Theta(p-x)$ with respect to x and use $\partial_x K(x-z) = -\partial_z K(x-z)$ to integrate by parts in z:

$$(\mathcal{A}\,\partial_x F_p)(x) + K(x-q)F_p(q) - K(x+q)F_p(-q) = -K(p-x). \tag{3}$$

Taking the $L^2[-q,q]$ inner product of (3) with the even density $\rho(x)$ and using $\langle \rho, \mathcal{A}\,\partial_x F_p\rangle = \int_{-q}^{q} \partial_x F_p(x)\,\mathrm{d}x = F_p(q) - F_p(-q)$ together with $\int_{-q}^{q} \rho(x)K(x\mp q)\,\mathrm{d}x = D(q) - 1 = t\rho(q) - 1$ gives

$$t\rho(q)\big[F_p(q) - F_p(-q)\big] = 1 - D(p). \tag{4}$$

Taking the inner product of (3) with the odd density $\eta(x)$ and using

$$\langle \eta, \mathcal{A}\,\partial_x F_p\rangle = \int_{-q}^{q} x\,\partial_x F_p(x)\,\mathrm{d}x = q\big[F_p(q) + F_p(-q)\big] - \big(P(p) - p\big)$$

together with $\int_{-q}^{q} \eta(x)K(x\mp q)\,\mathrm{d}x = \pm(V(q) - q) = \pm(t\eta(q) - q)$ cancels the bare $q[F_p(q)+F_p(-q)]$ boundary term and yields the dressed-energy endpoint identity **(c)**

$$t\eta(q)\big[F_p(q) + F_p(-q)\big] = P(p) - V(p). \tag{5}$$

Subtracting (4)–(5) at $p = q$ (where $f_p^{\pm} := F_p(\pm q) - F_q(\pm q)$ and $\overline{P}(p) := P(p) - P(q)$) confirms Eq. (12) of `frame_dictionary.tex`:

$$\mathfrak{M}_p := ta\rho(q)(f_p^+ - f_p^-) = -a\big[D(p) - D(q)\big], \qquad \mathfrak{Q}_p := \frac{f_p^+ + f_p^-}{2a\rho(q)} = \frac{\overline{P}(p) - V(p) + V(q)}{2at\rho(q)\eta(q)}. \tag{6}$$

A new exact opposite-edge dispersion identity (c) . Combining (4) multiplied by $v_s = \eta(q)/\rho(q) = V(q)/D(q)$ with (5) eliminates the right-edge backflow f_p^+ altogether and yields an exact finite-sea identity for the relative dispersion slope:

$$\boxed{E'(p) - v_s P'(p) = \overline{P}(p) - 2V(q)\big[F_p(-q) - F_q(-q)\big].} \tag{7}$$

Equation (7) shows that any deviation of $E'(p) - v_s P'(p)$ from $\overline{P}(p)$ comes exclusively from the opposite-edge backflow f_p^- (`audit_frame_dictionary.py`, Check 1).

Thermodynamic pressure response and exact quantum Jacobian (c) . Differentiating $(\mathcal{A}_q f)(x) = g(x)$ with respect to q at fixed $x \in (-q,q)$ and fixed h gives $(\mathcal{A}_q \partial_q f)(x) = \partial_q g(x) + K(x-q)f(q) + K(x+q)f(-q)$. For $f = \rho$ ($\partial_q g = 0$, $\rho(q) = \rho(-q)$), self-adjointness gives $\partial_q n = 2\rho(q) + \langle \rho, \mathcal{A}_q \partial_q \rho\rangle = 2t\rho(q)^2$ and $\partial_q P(p)|_{p,h} = t\rho(q)[F_p(q) + F_p(-q)]$. Since $a = 1/n(q)$, the chain rule $\partial_a = -a^{-2}(\partial_q n)^{-1}\partial_q$ confirms Eq. (13) of `frame_dictionary.tex`: $-a\partial_a P(p)|_{p,h} = [F_p(q) + F_p(-q)]/[2a\rho(q)]$. Likewise, differentiating the momentum-inversion relation $\overline{P}(q+hu(y)) = P_0(\kappa)$ ($\kappa = \sqrt{yh/b}$) with respect to y at fixed (h,a) using $\mathrm{d}P_0/\mathrm{d}y = (\mathrm{d}P_0/\mathrm{d}\kappa)(\mathrm{d}\kappa/\mathrm{d}y) = \frac{2h\sinh\kappa}{a}$ and $\mathrm{d}\overline{P}/\mathrm{d}y = hD(q+hu)\,\mathrm{d}u/\mathrm{d}y$ confirms Eq. (7) of `frame_dictionary.tex`:

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{2\sinh\kappa}{aD(q+hu)}, \qquad \frac{\mathrm{d}\overline{E}}{\mathrm{d}y} = \frac{2h\sinh\kappa}{a}\frac{V(q+hu)}{D(q+hu)}. \tag{8}$$

2.2 Independent 70-digit Fejér-I verification and missing Jacobian test

To test Eqs. (4)–(8) without reusing the package's Gauss–Legendre discretization or parameter point ($h = 0.4, q = 1.5$), Check 1 of `audit_frame_dictionary.py` implements a 70-digit `mpmath` Nyström solver based on Fejér's first rule (Chebyshev roots $z_k = \cos[\frac{(2k+1)\pi}{2N}]$) at ($h = 0.35, q = 1.4$) with $N = 68$ nodes ($a = 1.2645617476064018\ldots$). All endpoint, self-adjointness, opposite-edge (7), and fourth-order Richardson pressure-response identities pass with maximum residual 5.81×10^{-15}. **(c)**

Audit finding on test coverage (Point 5). Inspecting `frame_identities.py` and `frame_edge.py` reveals two test-coverage gaps in the package:

1. In `frame_edge.py:53`, `du_dy` is defined by evaluating the right-hand side $\frac{2\sinh\kappa}{D(p)}$ of (8) and never compared against the numerical y-derivative of the inverted coordinate $u(y) = (p(y) - q)/h$. Moreover, the second boxed equation in (8) for $\mathrm{d}\overline{E}/\mathrm{d}y$ is not tested in any of the three package scripts. **(c)** In `audit_frame_dictionary.py` (Check 1), we invert $\overline{P}(q + hu(y)) = P_0(\sqrt{yh/b})$ to 55 digits at $y \in \{0.5, 2.0\}$ and differentiate $u(y)$ and $\overline{E}(q + hu(y))$ directly with respect to y, confirming both equations in (8) to 1.95×10^{-25}. **(c)**

2. In `frame_identities.py:30--32`, the formal fixed-P chain rule (Eq. (9) of `frame_dictionary.tex`) is tested via `s.expand(E0p*(-P1/P0p)+E1-(E1-E0p*P1/P0p))==0` on unassigned scalar symbols—an algebraic tautology rather than a series inversion of $P_0(p) + hP_1(p)$. **(c)** Check 1 of `audit_frame_dictionary.py` replaces this with a symbolic Taylor series inversion of $P_0(p_0 + h\delta p_1 + h^2\delta p_2) + hP_1(p_0 + h\delta p_1)$, confirming Eq. (9). **(c)**

3 Point 2: Independent dispersion calculation and error-budget audit

3.1 Independent Fejér-I scan and rational Padé extrapolation

Check 2 of `audit_frame_dictionary.py` solves the finite-sea count equation $\int_{-q}^{q} \rho(x)\,\mathrm{d}x = 1$ ($a = 1$, $b = e^{-1/2}$) using the Fejér-I Chebyshev-root discretization on an independent geometric h-grid $h \in \{0.06, 0.03, 0.015, 0.0075\}$ with $N \in \{420, 640, 1050, 1800\}$ nodes, and extrapolates to $h = 0$ via four-point Neville polynomial extrapolation and rational Padé [2/1] approximants. Table 2 compares our independent Fejér-I extrapolations with Table 1 and Table 2 of `frame_dictionary.tex`. Every sampled entry agrees within the stated extrapolation spreads. **(c)**

y	$u - y$ (Fejér Nev)	$u - y$ (pkg)	u' (Fejér Nev)	u' (pkg)	$c_E(y)$ (Fejér Nev)	$c_E(y)$ (pkg)
0.1	−0.07635830	−0.0763583	0.35244896	0.3524490	0.07252661	0.0725266
1.0	−0.35748642	−0.3574864	0.84297699	0.8429770	0.05599866	0.0559987
4.0	−0.57571473	−0.5757147	0.96047441	0.9604744	0.05364428	0.0536443
16.0	−0.79553816	−0.7955379	0.99007018	0.9900702	0.05319386	0.0531937
64.0	−1.01595085	−1.0159485	0.99751500	0.9975150	0.05307429	0.0530723

Table 2: Independent Fejér-I dispersion extrapolations on $h \in \{0.06, 0.03, 0.015, 0.0075\}$ compared with Table 1 of `frame_dictionary.tex` (**(c)** , Check 2). For fixed unscaled $P \in \{0.15, 0.4, 0.6\}$ (Table 2), independent Fejér-I Neville extrapolations give $E_1(P) = 0.007775542, 0.020319169, 0.030098074$, agreeing with the package to $< 5 \times 10^{-9}$.

3.2 Three qualifications on the dispersion error budgets and closed forms

While the separation of mesh and edge-extrapolation diagnostics in `frame_dictionary.tex` is transparent, our audit identifies three substantive qualifications (`audit_frame_dictionary.py`, Check 2):

1. **Conditioning amplification vs. quadrature error at small y (c) :** At $y = 0.1$, the definition $c_{E,h}(y) = [\overline{E}(P_0) - E_0]/(hP_0)$ divides by $hP_0(\kappa) \propto h^{5/2}y^{3/2}$, which equals 9.57×10^{-8} at $h = 0.005$ (amplification factor $1/(hP_0) = 1.045 \times 10^7$). Consequently, IEEE 754 float64 subtraction roundoff and the root-finding tolerance (`xtol=6e-15` in `frame_edge.py:46`) are amplified into the reported mesh difference 4.62×10^{-10}. When multiplied by the ℓ^1 extrapolation weight factor 15.0 of the four-point $h \log h$ fit on geometric h-grids,

the package's own `mesh_propagated` diagnostic (6.94×10^{-9}) actually *exceeds* the reported `edge_spread` (5.86×10^{-9}) at $y = 0.1$.

2. **Exit from the small-amplitude regime at $y = 64$ (c) :** The expansion parameter governing finite-h lattice corrections at fixed y is $\kappa^2 = yh/b$. At $y = 64$, the four-point cubic and $h \log h$ fits include $h = 0.04$, where $\kappa = \sqrt{64 \times 0.04/e^{-1/2}} \simeq 2.054$ ($\kappa^2 \simeq 4.22 > 1$), well outside the small-κ regime.

3. **Exact analytical closed forms for $(v_s - b)/h$ and $c_E(y)$ asymptotics (c) :** Line 235 of `frame_dictionary.tex` states that the extrapolated sound-velocity shift $(v_s - b)/h \simeq 0.0795774724$ is "a numerical estimate, not an independently proved closed form." In fact, for $a = 1$ it is the exact one-loop Fermi-velocity shift

$$\lim_{h\to 0} \frac{v_s(h) - b}{h} = \frac{1}{4\pi} = 0.07957747154594767\ldots, \tag{9}$$

which agrees with the package's cubic extrapolation (0.079577471578) to 3.2×10^{-11} and with our independent Fejér-I Neville extrapolation (0.079577471692) to 1.46×10^{-10}. Moreover, since $\overline{E}(p)/P_0(\kappa) \to v_s$ as $y \to 0^+$ while integration of the bulk dispersion relation $\mathrm{d}\overline{E}/\mathrm{d}u = v_s \mathrm{d}\overline{P}/\mathrm{d}u + \frac{\sqrt{b}}{3} h^{5/2} y^{3/2} + O(h^{7/2})$ against $\mathrm{d}u = [1 - \frac{1}{2\pi y} - \frac{d_0}{y^2} + O(y^{-3})]\,\mathrm{d}y$ subtracts $\frac{1}{12\pi}$ at $y \to \infty$, $c_E(y)$ interpolates monotonically between exact analytical endpoints:

$$\lim_{y\to 0^+} c_E(y) = \frac{1}{4\pi} \simeq 0.07957747, \qquad c_E(y) = \frac{1}{6\pi} - \frac{d_0}{2y} + O(y^{-2}) \quad (y \to \infty), \tag{10}$$

where $\frac{1}{6\pi} \simeq 0.0530516477$ and $-\frac{d_0}{2} = \frac{1}{8\pi^2} - \frac{1}{96} \simeq +0.00224848$. At $y = 16$, Eq. (10) gives 0.05319218, matching Table 1 (0.0531937) to 1.5×10^{-6}. **(c)**

4 Point 3: The mixed classical-frame remainder and its exact reduction

4.1 Exact reduction of the edge limit to the one-body Jacobian

In Section 5 of `frame_dictionary.tex`, the symmetric classical-frame remainder at fixed scaled dressed momenta is

$$S_h(y, z) = \frac{\mathfrak{M}_y \mathfrak{Q}_z + \mathfrak{M}_z \mathfrak{Q}_y - C_{\mathrm{cl}}(y, z) - C_{\mathrm{cl}}(z, y)}{h}, \qquad C_{\mathrm{cl}}(y, z) = -2\kappa_y \big[P_0(\kappa_z) - 2b \sinh \kappa_z\big].$$

Using the exact reciprocity identity $\mathfrak{M}_y \mathfrak{Q}_z + \mathfrak{M}_z \mathfrak{Q}_y = t(f_y^+ f_z^+ - f_y^- f_z^-)$ and (4), we have identically at every finite h:

$$f_y^+ - f_y^- = 1 - \frac{D(q + hu(y))}{D(q)}.$$

As $h \to 0$ at fixed $y > 0$, the opposite-edge response scales as $f_y^- = O(h) \to 0$, while $D(q) = \sqrt{\pi h/b}\,[1 + O(h)]$ ($a = 1$) and $D(q + hu(y)) = 2\sqrt{yh/b}/u'(y)\,[1 + O(yh)]$. Hence $\lim_{h\to 0} f_y^+ = 1 - \frac{2\sqrt{y}}{\sqrt{\pi}\,u'(y)}$, and since $[C_{\mathrm{cl}}(y, z) + C_{\mathrm{cl}}(z, y)]/h \to 8\sqrt{yz}$, the exact $h \to 0$ limit of $S_h(y, z)$ is **(c)** (`audit_frame_dictionary.py`, Check 3):

$$\boxed{S_0(y, z) := \lim_{h\to 0} S_h(y, z) = 8\sqrt{yz}\left(\frac{1}{u'(y)\,u'(z)} - 1\right) - 4\sqrt{\pi}\left(\frac{\sqrt{y}}{u'(y)} + \frac{\sqrt{z}}{u'(z)}\right) + 2\pi.} \tag{11}$$

Equation (11) establishes two decisive facts:

1. **Confirmation of `frame_dictionary.tex` lines 339–346 (c) :** For a single pair (y, z), the $-D(q)$ endpoint subtraction contributes the additive one-body terms $-4\sqrt{\pi}\big(\frac{\sqrt{y}}{u'(y)} + \frac{\sqrt{z}}{u'(z)}\big) + 2\pi$ in (11). At $(y, z) = (4, 1)$, keeping only the first term of (11) gives $+3.7614$, whereas the full formula (11) gives -13.1290, matching our extrapolated finite-sea value -13.1290 to $< 10^{-6}$.

2. **Exact cancellation of endpoint terms in any mixed rectangle (c) :** In any rectangular second difference $\mathcal{D}_0(y_1, y_0; z_1, z_0) = S_0(y_1, z_1) - S_0(y_0, z_1) - S_0(y_1, z_0) + S_0(y_0, z_0)$, the additive one-body terms $-4\sqrt{\pi}\big(\frac{\sqrt{y}}{u'(y)} + \frac{\sqrt{z}}{u'(z)}\big) + 2\pi$ cancel identically, and the remaining bilinear factorises:

$$\boxed{\mathcal{D}_0(y_1, y_0; z_1, z_0) = 8\left(\frac{\sqrt{y_1}}{u'(y_1)} - \frac{\sqrt{y_0}}{u'(y_0)}\right)\left(\frac{\sqrt{z_1}}{u'(z_1)} - \frac{\sqrt{z_0}}{u'(z_0)}\right) - 8\big(\sqrt{y_1} - \sqrt{y_0}\big)\big(\sqrt{z_1} - \sqrt{z_0}\big).} \tag{12}$$

   Thus the surviving mixed rectangle in Table 3 of `frame_dictionary.tex` is governed *solely* by the one-body quantum Jacobian $u'(y) \neq 1$: the classical frame subtraction $C_{\mathrm{cl}}(y, z)$ used the bare classical factor $\sqrt{y}$ rather than the Jacobian-dressed factor $\sqrt{y}/u'(y)$. Substituting $u'(y) = 1 - \frac{1}{2\pi y} - \frac{d_0}{y^2} + O(y^{-3})$ into (12) for $(y_1, y_0; z_1, z_0) = (2Y, Y; Y/2, Y/4)$ proves analytically that $\mathcal{D}_0(Y)$ has a pure $1/Y$ expansion (no $1/\sqrt{Y}$ term) with exact nonzero large-Y limit **(c)**

$$\lim_{Y\to\infty} \mathcal{D}_0(Y) = -\frac{15\sqrt{2} - 20}{\pi} = -0.38617464750248\ldots \tag{13}$$

4.2 Alternative extrapolation bases and new point sets

Table 3 tests $\mathcal{D}_h(Y)$ across five extrapolation bases not used in `frame_analysis.py` (Padé [1/1], Padé [2/1], three-point $1 + h + h\log h$, four-point $1 + h + h^{3/2} + h^2$, and four-point $1 + h + h^2 + h^2\log h$) and compares them with the analytical reduction (12). Check 3 of `audit_frame_dictionary.py` also evaluates (12) on three new independent rectangles using the Fejér-I solver and directly solved four-term phases $\widehat{\Phi}(1, 2) + \widehat{\Phi}(2, 1)$: $(16, 4) \times (1, 0.25)$ gives -3.97463 (Eq. (12): -3.97462), $(8, 2) \times (0.5, 0.1)$ gives -3.26615 (Eq. (12): -3.26615), and $(16, 4) \times (2.56, 1.44)$ gives -0.79218 (Eq. (12): -0.79218). **(c)** In every basis and point set, the mixed classical-frame remainder survives the edge limit. Furthermore, the symbolic two-term $1/Y$ expansion of (12) matches $\mathcal{D}_0(32)$ to 2.86×10^{-5}, and fitting $\mathcal{D}_0(Y)$ on $Y \in \{8, 16, 32\}$ to a quadratic in $1/Y$ (or on $Y \in \{4, 8, 16, 32\}$ to a cubic in $1/Y$, without the spurious $1/\sqrt{Y}$ term that pulled the package's first intercept to -0.3276) gives -0.38652 (and -0.38624), within 3.4×10^{-4} of the exact asymptote (13). **(c)**

Y	Pkg quad	Pkg cubic	Padé [1/1]	Padé [2/1]	$h^{3/2}$ fit	$h^2 \log h$ fit	Exact Eq. (12)
4	−0.504362	−0.504365	−0.504377	−0.504365	−0.504372	−0.504368	−0.504365
8	−0.434983	−0.435004	−0.435089	−0.435003	−0.435065	−0.435033	−0.435003
16	−0.408705	−0.408890	−0.409478	−0.408878	−0.409412	−0.409136	−0.408877
32	−0.395752	−0.397374	−0.401151	−0.397163	−0.401961	−0.399536	−0.397225

Table 3: Survival of the mixed classical-frame rectangle $\mathcal{D}_0(Y)$ across alternative extrapolation bases and comparison with the exact one-body Jacobian reduction (12) (**(c)** , Check 3).

5 Point 4: Audit of the claimed antisymmetric two-body freedom

In Section 6 of `frame_dictionary.tex`, the authors exhibit

$$A_\lambda(y_1, y_2) = \lambda \frac{y_1 y_2 (y_1 - y_2)}{(y_1 + y_2)^3}, \tag{14}$$

and note that shifting $B(p_1, p_2) = \widehat{\Phi}(p_1, p_2) - \mathfrak{M}_{p_1}\mathfrak{Q}_{p_2}$ to $B + hA_\lambda$ preserves all one-body observables, axis vanishing $A_\lambda(y_1, 0) = 0$, the tree-level action limit, and the symmetric reciprocity sum $(B + hA_\lambda)(1, 2) + (B + hA_\lambda)(2, 1) = 0$, while shifting B/h at $O(h^0)$ by $A_\lambda(Y, r^2 Y) = \lambda r^2(1 - r^2)/(1 + r^2)^3$.

Is A_λ a physical ambiguity or an artefact of the tested constraints? Check 4 of `audit_frame_dictionary.py` confirms the algebraic properties of A_λ **(c)** , but proves that it is **only an artefact of the incomplete set of constraints tested by the package**, not a genuine ambiguity of the physical Toda/Boussinesq problem:

1. **Spatial cluster decomposition and Darboux frame separability (excludes A_λ) (c) :** In the asymptotic two-soliton sector $|X_1 - X_2| \to \infty$, local charge densities cluster into single-soliton contributions, so the symplectic cross-term $\sum_{i<j} \mathrm{d}\Pi_i \wedge \mathrm{d}M_j$ and the resulting Darboux coordinate transformation $X_j \mapsto X'_j$ are generated by a finite bilinear $\sum_{a=1}^{K} f_a(y_1) g_a(y_2)$ of one-body state functions (Schmidt rank ≤ 2). Consequently, any frame correction to the two-body phase must have vanishing 3×3 minors $\det_{1 \leq i,j \leq 3} F(y_i, z_j) = 0$. By contrast, $A_\lambda(y_1, y_2)$ has infinite Schmidt rank: at $(y_1, y_2) \in \{1, 2, 3\} \times \{4, 5, 6\}$, Check 4 verifies symbolically that $\det_{1 \leq i,j \leq 3} A_1(y_i, z_j) = -\frac{18232351733}{8133427676989440} \neq 0$. Thus hA_λ cannot arise from any redefinition of one-body soliton charges or Darboux coordinates.

2. **Microscopic Toda / Krein–Wiener–Hopf differential equation (excludes A_λ) (c) :** The package tested only the symmetric sum $\widehat{\Phi}(1, 2) + \widehat{\Phi}(2, 1)$, whose kernel trivially contains every antisymmetric two-body function. However, the Toda Fredholm equation fixes $\widehat{\Phi}(p_1, p_2)$ itself: in the Fermi-edge limit, $(\partial_{u_1} + \partial_{u_2})\widehat{\Phi}_q(u_1, u_2) = 2\psi_-(u_1)[\widetilde{\Psi}_-(u_2) - \widetilde{\Psi}_-(0)]$ is a rank-1 separable product of one-body functions, which uniquely determines $\widehat{\Phi}_q(u_1, u_2)$ from the boundary condition $\widehat{\Phi}_q(u_1, 0) = 0$. Applying $(\partial_{y_1} + \partial_{y_2})$ to $A_1(y_1, y_2)$ gives $+(y_1 - y_2)(y_1^2 - 4y_1 y_2 + y_2^2)/(y_1 + y_2)^4$, whose 2×2 separability determinant at $(1, 3) \times (2, 5)$ is $\frac{277}{1920000} \neq 0$.

3. **Crossing / reflection parity and rapidity analyticity (excludes A_λ) (c) :** With $y_j = \beta_j^2/\hbar_B$ and $r = \beta_2/\beta_1$, $A_\lambda(Y, r^2 Y) = \lambda r^2(1 - r^2)/(1 + r^2)^3$ is *even* under $r \mapsto -r$ ($\beta_2 \mapsto -\beta_2$), whereas the physical one-loop residual $R_0(\infty, r) = -\frac{4}{\pi}(\chi_2(r) - r)$ is strictly *odd* in r. Furthermore, $A_\lambda(Y, r^2 Y)$ introduces unphysical third-order poles at $r = \pm i$ ($\beta_1^2 + \beta_2^2 = 0$), where the Toda Bethe kernel is regular.

4. **What does *not* remove A_λ (c) :** The tree-level $O(\hbar^{-1})$ classical limit does not remove hA_λ because $hA_\lambda/h = O(\hbar^0)$ is one-loop order. Likewise, scalar three-body Yang–Baxter and three-body Bethe momentum conservation $\sum_{j=1}^{3} P(p_j) L = 2\pi h \sum_{j=1}^{3} I_j$ hold for any antisymmetric pair phase, though three-body linear sea backflow $F_{p_2,p_3} = F_{p_2} + F_{p_3}$ combined with one-body charge separability excludes non-separable frame terms.

Remaining physical open question (d) . While A_λ is an artefact of testing only symmetric reciprocity, a narrower rank-2 question remains genuinely open **(d)** : identifying the unique antisymmetric combination of *one-body* endpoint responses $(\mathfrak{M}_1\mathfrak{Q}_2 - \mathfrak{M}_2\mathfrak{Q}_1)$ with the canonical collective-coordinate operators in the continuum Boussinesq Hilbert space still requires the projected fluctuation determinant and zero-mode measure.

6 Additional checks, verdict table, and failures recorded

Continuum mode-sum non-uniformity and Phase 2i factor product (Check 5) (c) . Check 5 of `audit_frame_dictionary.py` confirms Eq. (20) of `frame_dictionary.tex`: for $\Delta P_g = \epsilon\alpha(\log\alpha + c)/\pi$, the relative shift at fixed $y = \alpha^2/\hbar_B$ is $\hbar_B \Delta P_g / P_{g,0} = \frac{3}{2\pi y}(\log\alpha + c) + O(\alpha^2 \log\alpha)$, which is non-uniform as $\alpha \to 0$ at fixed y. Check 5 also verifies to 70 digits that for $\Phi_{\mathrm{WH}}(s) = \sqrt{2\pi/s}\,\Gamma(1 + \frac{s}{2\pi}) \exp(-\frac{s}{2\pi} \ln \frac{s}{2\pi e})$,

$$\Phi_{\mathrm{WH}}(i\omega)\Phi_{\mathrm{WH}}(-i\omega) = \frac{\pi e^{|\omega|/2}}{\sinh(|\omega|/2)} \qquad (\omega \neq 0),$$

confirming that the middle display $\frac{2\pi^2 e^{\pi|\omega|}}{\sinh(\pi|\omega|)}$ in Eq. (3) of `toda_wh_and_kp_synthesis.tex` contains a typographical rescaling ($\omega \mapsto 2\pi\omega$ and an extra 2π factor). **Qualification (c) :** Because Eq. (4) of `toda_wh_and_kp_synthesis.tex` divides by $\sqrt{2\pi}$ to define $\widetilde{\Phi}_{\mathrm{WH}}(s) = \Phi_{\mathrm{WH}}(s)/\sqrt{2\pi}$ and uses only $\widetilde{\Phi}_{\mathrm{WH}}(s)$ in its Laplace transforms, the Eq. (3) display typo does not propagate into the Phase 2i expansion of $\delta u(y)$ or $u'(y) - 1$, which agree with Table 1 of `frame_dictionary.tex` to 5.5×10^{-7} (δu) and 2.1×10^{-7} ($u' - 1$) at $y = 8$, and 9.9×10^{-8} (δu) and 5.7×10^{-9} ($u' - 1$) at $y = 16$.

Failures recorded.

1. Running `frame_identities.py` and `frame_analysis.py` directly overwrites `frame_identities.json` and `frame_analysis.json` with fresh `seconds` timestamps, modifying tracked files outside the `audit_frame_dictionary` namespace. Both JSON files were restored byte-for-byte to their initial SHA-256 hashes, and `audit_frame_dictionary.py` enforces an exact SHA-256 invariance guard across all seven audited files.

2. In an initial mental calculation of the large-Y limit of (12), the factor $\frac{1}{\sqrt{Y/2}} - \frac{1}{\sqrt{Y/4}} = -\frac{\sqrt{2}(\sqrt{2}-1)}{\sqrt{Y}}$ was written without the factor of $\sqrt{2}$, giving $-\frac{22-15\sqrt{2}}{\pi} \simeq -0.2504$ instead of the exact limit $-\frac{15\sqrt{2}-20}{\pi} \simeq -0.386175$. Exact SymPy series expansion in Check 3 of `audit_frame_dictionary.py` caught and corrected the arithmetic before any check was finalised; no tolerance was weakened.

3. In Check 1 of `audit_frame_dictionary.py`, an initial $N = 56$ Fejér-I (Chebyshev-root) mesh at $(h = 0.35, q = 1.4)$ gave maximum endpoint/pressure error 2.06×10^{-12}, exceeding our strict 10^{-12} assertion threshold because N-node Fejér-I quadrature has polynomial degree of exactness $N - 1$ (half that of N-node Gauss–Legendre quadrature). Refining the Fejér-I mesh to $N = 68$ passed the unchanged 10^{-12} threshold without weakening any assertion tolerance.

4. In Check 3 of `audit_frame_dictionary.py`, a four-point quadratic $1/Y$ fit across $Y \in \{4, 8, 16, 32\}$ initially failed a 5×10^{-4} assertion against $\mathcal{D}_0(\infty)$ (giving -0.3875 vs. -0.3862) because at $Y = 4$ the lowest rectangle corner is $z_0 = Y/4 = 1$, where $1/z_0 = 1$ is $O(1)$ and contributes $O(10^{-3})$ higher-order $1/z^3$ terms. Replacing the heuristic fit by the exact symbolic $O(1/Y)$ and $O(1/Y^2)$ coefficients of (12) (which match $\mathcal{D}_0(32)$ to 2.86×10^{-5}) and testing the quadratic fit on $Y \in \{8, 16, 32\}$ alongside the cubic $1/Y$ fit on $Y \in \{4, 8, 16, 32\}$ resolved the issue cleanly.

5. In Check 4 of `audit_frame_dictionary.py`, the symbolic assertion on $(\partial_{y_1} + \partial_{y_2})A_1(y_1, y_2)$ initially had the opposite overall sign ($-$ instead of $+$ in $+(y_1 - y_2)(y_1^2 - 4y_1y_2 + y_2^2)/(y_1 + y_2)^4$), `sp.solve` for the poles $r = \pm i$ of $(1 + r^2)^3$ was initially called on a symbol declared with `positive=True` (which filtered out complex roots), and the three-body pairwise sum initially used sequential `.subs({y1:y2, y2:y1})` instead of simultaneous `.xreplace`; all

three were fixed in Check 4 without affecting the 2×2 Krein separability determinant ($\frac{277}{1920000} \neq 0$).

6. The package's claims that $(v_s - b)/h$ has no proved closed form, that $\mathcal{D}_0(Y)$ may require $1/\sqrt{Y}$ terms, that $c_E(0.1)$ is purely edge-spread-limited, and that $A_\lambda(y_1, y_2)$ represents an unremovable two-body physical freedom fail the analytical and physical tests documented in Sections 3–5 and are recorded as assertions in `audit_frame_dictionary.py`.

Audited claim in `frame_dictionary.tex`	Verdict	Independent evidence (`audit_frame_dictionary.py`)
Classical Toda–Boussinesq charge and velocity scalings, Eqs. (1), (8) **(c)**	**Confirmed (c)**	Check 1: exact SymPy limits and series
Finite-sea dressed-momentum and dressed-energy endpoint identities, Eqs. (2)–(5) **(c)**	**Confirmed (c)**	Check 1: integration-by-parts proof; 70-dps Fejér-I solver ($< 5.8 \times 10^{-15}$); new identity (7)
Exact quantum momentum and energy Jacobians, Eqs. (6)–(7) **(c)**	**Confirmed** (coverage qualified) **(c)**	Check 1: direct 70-dps y-derivative check ($< 2.0 \times 10^{-25}$); package omitted $\mathrm{d}\overline{E}/\mathrm{d}y$ test
Formal fixed-P one-loop energy inversion, Eq. (9) **(c)**	**Confirmed** (test qualified) **(c)**	Check 1: non-tautological series proof (replacing tautology in `frame_identities.py:30--32`)
Tabulated dispersion $u - y$, u', $c_E(y)$, $E_1(P)$, and $(v_s - b)/h$, Tables 1–2 **(c)**	**Qualified (c)**	Check 2: central values reproduced via Fejér-I; at $y = 0.1$, propagated mesh error > edge spread due to $1/(hP_0)$ amplification; at $y = 64$, $\kappa(0.04) \simeq 2.05$; exact $(v_s - b)/h = 1/(4\pi)$ and $c_E(\infty) = 1/(6\pi)$ proved
Thermodynamic pressure response and endpoint derivatives, Eqs. (11)–(13) **(c)**	**Confirmed (c)**	Check 1: analytic proof and 4th-order Richardson q-differences ($< 5.8 \times 10^{-15}$)
Four-term reciprocity and necessity of $-D(q)$ endpoint subtraction, Eqs. (15)–(16) **(c)**	**Confirmed (c)**	Check 3: Eq. (11) proves dropping $-D(q)$ shifts $S_0(4,1)$ from -13.129 to $+3.761$
Survival of mixed classical-frame rectangle $\mathcal{D}_h(Y)$ as $h \to 0$, Table 3 **(c)** /**(d)**	**Confirmed** (limit qualified) **(c)**	Check 3: confirmed across 5 new bases and 3 new point sets; exact reduction (12) to $u'(y)$ proves $\mathcal{D}_0(\infty) = -(15\sqrt{2} - 20)/\pi \simeq -0.386175$ with no $1/\sqrt{Y}$ term
Two-body antisymmetric freedom $A_\lambda(y_1, y_2)$, Eqs. (18)–(19) **(c)** /**(d)**	**Qualified / Refuted as physical (c)**	Check 4: valid counterexample to reciprocity alone, but refuted as a physical frame ambiguity by cluster separability ($\det_{3\times 3} \neq 0$), Krein PDE, and r-parity/poles
Continuum mode-sum non-uniformity Eq. (20) and Phase 2i WH product Eq. (21) **(c)**	**Confirmed** (scope qualified) **(c)**	Check 5: product $\pi e^{\lvert\omega\rvert/2}/\sinh(\lvert\omega\rvert/2)$ verified to $< 10^{-65}$; downstream $\widetilde{\Phi}_{\mathrm{WH}}$ unaffected

Table 4: Summary verdict table for the `frame_dictionary` package.